The Northern Lights

Natural light display in Earth's polar regions

The aurora borealis, commonly known as the northern lights, is a natural light display in Earth's sky, predominantly seen in high-latitude regions around the Arctic and Antarctic. Auroras display dynamic patterns of brilliant lights that appear as curtains, rays, spirals, or dynamic flickers covering the entire sky. Auroras are caused by disturbances in Earth's magnetosphere caused by solar wind. These disturbances alter the trajectories of charged particles in the magnetospheric plasma. The particles, mainly electrons and protons, precipitate into the upper atmosphere, causing ionization and excitation of atmospheric constituents, which produces optical emissions. Different gases in Earth's atmosphere produce different colors in the aurora. Oxygen molecules produce green and red auroras, the most common colors. Nitrogen molecules create blue and purple hues. The altitude at which collisions occur determines the color—green auroras typically occur at altitudes of 100-300 kilometers. The aurora borealis has fascinated humans throughout history. Ancient Norse mythology attributed the lights to the Bifrost Bridge, a burning rainbow bridge that connected Earth to Asgard, the realm of the gods. Medieval Europeans thought auroras were